Poisson Random Variables

$$Y \sim Poisson(\lambda)$$

$E[Y] = \lambda$ $\qquad\qquad$ $Var[Y] = \lambda$

- Y = count per some unit of effort
- Y can take on values 0,1,2,3,... (*unbounded count*)
- λ = mean count per unit effort in the population

> **Note**
>
> The Sleuth uses μ_i instead of λ_i, but λ_i is more typical

Examples

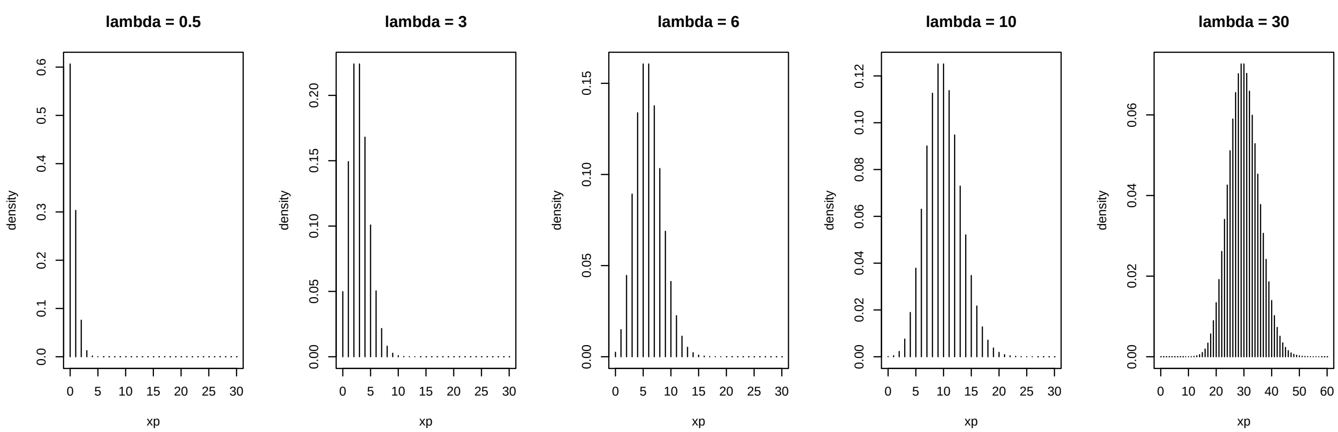
Identify the "count", "unit of effort" and the "sample unit" for the following Poisson RVs
Y = # of traffic accidents at an intersection over a year

Y = # shrimp caught in 1-km tow of a net event

Y = # of flaws on a computer chip

Poisson Distribution

$$Pr[Y = y] = \frac{e^{\lambda}\lambda^{y}}{y!}, \;\; y = 0, 1, 2, 3, 4, \ldots$$



Poisson GLM

Elephant case study

Larger male elephants tend to be more successful in mating (can fight off smaller male elephants), and elephants grow throughout their lifetime. In a population of African elephants in Amboseli NP, Kenya, Joyce Pool collected data on 41 different male elephants.

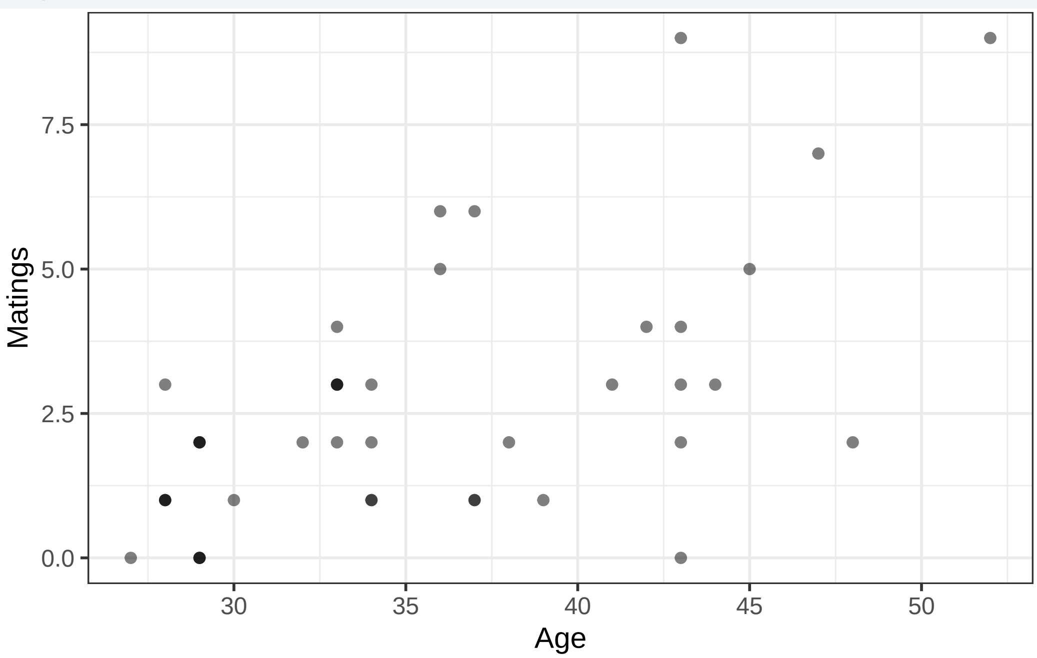
- `Age` = age of elephant at the b
- `Matings` = total number of successful matings

Data are in `Sleuth3::case2201`

```r
library(tidyverse)
library(Sleuth3)
case2201 |>
  head(n = 10)
```

```
   Age Matings
1   27       0
2   28       1
3   28       1
4   28       1
5   28       3
6   29       0
7   29       0
8   29       0
9   29       2
10  29       2
```

```r
case2201 |>
  ggplot(aes(x = Age, y = Matings)) +
  geom_point(alpha = 0.5) + theme_bw()
```



RQ

What is the relationship between mating success and age? *Do males have diminished success after some optimal age?*

Specify a model to address this RQ. Include distribution of Y, linear predictor, and link function.

Estimation

`glm` uses ML estimation to estimate $\beta_0, \beta_1, \beta_2$

```
fit_quad <- glm(Matings ~ poly(Age,2), data = case2201, family = "poisson")
summary(fit_quad)
```

```

Call:
glm(formula = Matings ~ poly(Age, 2), family = "poisson", data = case2201)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)     0.8759     0.1060   8.265  < 2e-16
poly(Age, 2)1   2.9690     0.6375   4.657 3.21e-06
poly(Age, 2)2  -0.2347     0.5495  -0.427    0.669

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 75.372  on 40  degrees of freedom
Residual deviance: 50.826  on 38  degrees of freedom
AIC: 158.27

Number of Fisher Scoring iterations: 5
```

Estimated model

```
summary(fit_quad)$coef
```

Deviance GOF test

Informal assessment of adequacy of model fit is same in Poisson as Binomial GLM. H_0: model is adequate vs. H_a : model is inadequate.

- Deviance stat = `Residual deviance`, follows $\sim \chi^2_{n-p}$ under H_0

```
Residual deviance: 50.826  on 38  degrees of freedom
```

> **Caution**
>
> This test is not trustworthy when Poisson means are small (< 5)

Deviance GOF test

Should we trust the χ^2_{38} distribution as a reasonable approximation for the null distribution in this case?

```r
fitted(fit_quad)
```

```
       1        2        3        4        5        6        7        8 
1.205366 1.317143 1.317143 1.317143 1.317143 1.436813 1.436813 1.436813 
       9       10       11       12       13       14       15       16 
1.436813 1.436813 1.436813 1.564664 1.845963 1.999881 1.999881 1.999881 
      17       18       19       20       21       22       23       24 
1.999881 1.999881 2.162911 2.162911 2.162911 2.162911 2.516912 2.516912 
      25       26       27       28       29       30       31       32 
2.708089 2.708089 2.708089 2.908783 3.118983 3.567606 3.805739 4.052794 
      33       34       35       36       37       38       39       40 
4.052794 4.052794 4.052794 4.052794 4.308474 4.572418 5.123315 5.409210 
      41 
6.606923 
```

Generally, what does a large $p - value$ from an *omnibus GOF* test mean?

Residuals

Deviance Residuals: measure of how different the max value of the log likelihood is at observation i - the model estimated value of the log-likelihood at observation i (Sleuth for formula)
Pearson Residuals: observed response minus its estimated mean divided by its estimated standard deviation

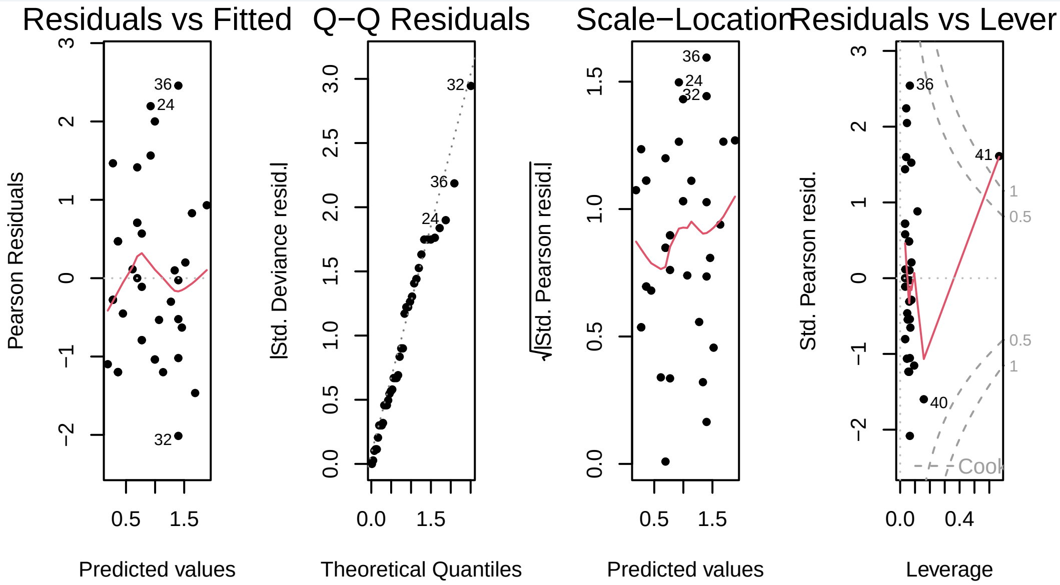
- Both are approximately $N(0, 1)$ when λ is large enough.

Residual diagnostics

Note we do not have large Poisson means in this example…

```r
#|echo: true
#| fig-width: 12
#| fig-height: 4
par(mfrow = c(1,4))
```

```r
# plotting Pearson resids
plot(fit_quad, pch = 20)
```



Back to the RQ

Do males have diminished success after some optimal age?

```r
round(summary(fit_quad)$coef, digits = 4)
```

```
               Estimate Std. Error z value Pr(>|z|)
(Intercept)      0.8759     0.1060  8.2647   0.0000
poly(Age, 2)1    2.9690     0.6375  4.6570   0.0000
poly(Age, 2)2   -0.2347     0.5495 -0.4271   0.6693
```

Refine model

What is the relationship between mating success and age?

```r
fit_linear <- glm(Matings ~ Age, data = case2201, family = "poisson")
summary(fit_linear)
```

```

Call:
glm(formula = Matings ~ Age, family = "poisson", data = case2201)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.58201    0.54462  -2.905  0.00368
Age          0.06869    0.01375   4.997 5.81e-07
```

```
(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 75.372  on 40  degrees of freedom
Residual deviance: 51.012  on 39  degrees of freedom
AIC: 156.46

Number of Fisher Scoring iterations: 5
```

Summarise inferential model

1. Write the estimated model
2. Address the research question by interpreting the estimate of the coefficient of interest and summarizing statistical findings on the original scale (include a test and CI in your summary). You can ignore overdispersion.
    - You'll need to construct a Wald-based 95% CI!

Quasipoisson for overdispersion

Quasipoisson
$\mu\{Y_i|X_i\} = \lambda_i$ $Var\{Y_i|X_i\} = \psi^* \lambda_i$
Which model should we use to estimate the degree of overdispersion in the elephant example? `fit_linear` or `fit_quad`?

Overdispersion estimate

From `fit_quad`

```
Residual deviance: 50.826  on 38  degrees of freedom
```

$\widehat{\psi} =$

```r
sum_pearson2 <- sum(resid(fit_quad, type = "pearson")^2)
psi_hat_star <- sum_pearson2/38
psi_hat_star
```

```
[1] 1.175194
```

$\widehat{\psi} =$

Checking for OD in general

1. Is extra Poisson variation likely?
2. Plot sample variance vs. sample mean for groups of responses with the same x values
3. Did deviance GOF from rich model result in evidence of lack-of-fit?
4. Conduct residual analysis

Accounting for OD

Same as for Binomial GLM!